

AQUARIUM JELLYFISH · TANK SWIM



EXO-MEDUSA

Consumer kit v6 · Bambu Lab A1

Ø 280 × 380 mm · phone is the remote · no steering
360° servo drive · eight TPU paddles · dry capsule
First tank swim for a Bambu Lab A1 owner — Thailand, or anywhere

2

weekends

8

tentacles

GPIO4

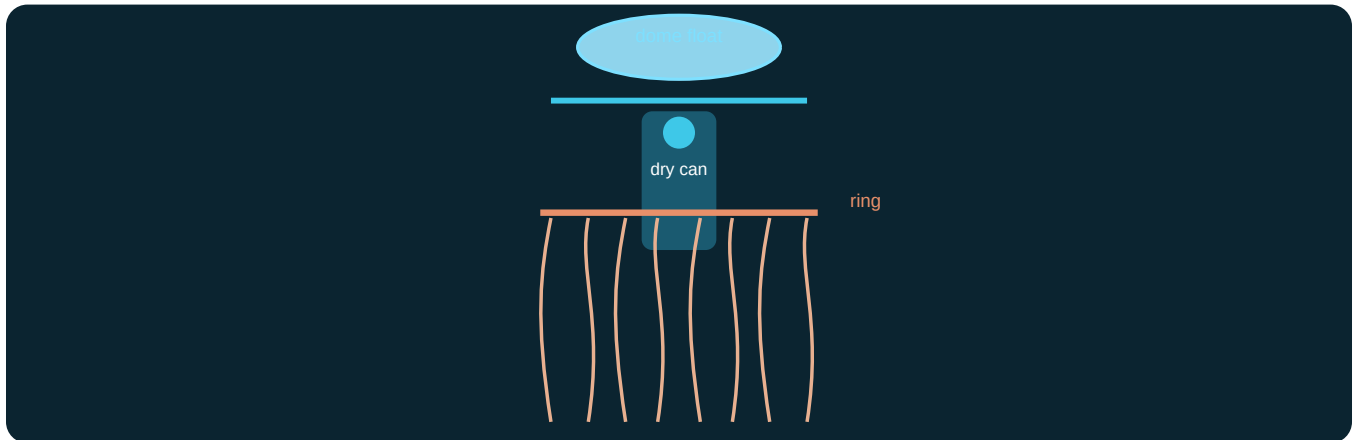
leak pin

v6

firmware

What you are building

You are building EXO-MEDUSA: an aquarium jellyfish about $\varnothing 280 \times 380$ mm. A dry electronics capsule sits in the middle. A 360° servo turns magnets inside the can. Matching magnets outside follow through the wall. A 3-lobe cam then pumps a sliding ring 10 mm, three times per turn. The ring bends eight tentacles inward together. That pulse is the swim.



Mesh preview of the idea — not a photo of a finished robot.

- Your phone browser is the remote. No app to install.
- There is no steering. It pulses and drifts. That is this kit.
- No electrical part lives in the water. Servo, battery and ESP32 stay dry.
- Firmware is v6. Leak sense is on GPIO4 (not GPIO5 from the old French notice).
- Drive is a 360° continuous servo: DS04-NFC or MG996R 360. Not a Furitek.

Tentacles in this kit

Print the TPU paddle eight times (original zip file 21, 22 × 209 mm). Kit path: `print/stl/optional/legacy_21_flanc_TPU_NOT_piece19.stl`. It is a paddle, not the missing Fin-Ray ladder. It still fits the 8.6 × 3.0 mm slots. Print 20_tentacle_bar_PETG_x8.stl eight times (alias 20_barrette_PETG_x8.stl). Cut eight films from GABARIT_FLANCS_1-1.pdf at 100 % scale. Skip 19_tentacle_skeleton_TPU_x8.stl for these steps — different mesh.

Budget two weekends. One PH1 screwdriver. Super glue. A Bambu Lab A1. Thailand shop list uses Lazada / Shopee search terms. The same parts exist anywhere.

Safety — read this first

N52 magnets can kill if swallowed

They pinch hard and chip if they slam together. Keep them away from children. A swallowed N52 magnet is lethal. Two magnets (or one magnet and metal) can trap bowel. This is not a toy drawer.

18650 shorts

Never short the + and – tabs. Never puncture. Charge only through the TP4056 module. No bare-cell chargers. No loose cells in a pocket with keys.

CA glue

Cyanoacrylate bonds skin in about five seconds. Ventilate. Keep it off eyes. Have acetone or a CA debonder nearby.

Servo 5 V does not go through the ESP32

The servo red wire goes to the boost 5 V output. The ESP32 may share that rail. Do not feed servo current through the chip. A 470 μ F capacitor sits across servo 5 V (long leg on +). Missing that cap is why a servo twitches and will not turn.

Aquarium silicone — no fungicide

100 % clear aquarium silicone. No fungicide. No anti-mould. Bathroom silicone can kill fish and is the wrong tube.

Also. Never force the bayonet lid — a wire is on the O-ring. Coat every wet-side magnet (outer rotor) with nail varnish or thin epoxy. Nickel plating is not enough in water. If the float dome leaks, the robot inverts. The dome is the float, not decoration.

What you need

Buy on Lazada / Shopee Thailand, or anywhere. Prices are ballpark THB from the kit shop list, except the servo (the old spreadsheet had the wrong motor).

Electronics

What	Qty	Search	~THB
360° servo DS04-NFC or MG996R 360	1	DS04-NFC 360 / MG996R 360 continuous	—
ESP32-C3 SuperMini	1	ESP32-C3 SuperMini	105
18650 + holder (3000 mAh)	1	18650 battery holder	140
TP4056 with protection (DW01)	1	TP4056 protection	35
5 V boost ≥ 2 A (MT3608)	1	MT3608 boost 5V	35
470 μ F 10 V electrolytic	1	470uF electrolytic	17.5
SS49E linear Hall	1	SS49E hall sensor	17.5
NTC 10k B3950	1	NTC 10k 3950	17.5
Reed switch + small magnet	1	reed switch glass	35
Resistors 10k $\times$ 1, 100k $\times$ 2, 470k $\times$ 1	1 kit	resistor kit	35

Mechanics, film, glue

What	Qty	Search	~THB
N52 magnets 8 $\times$ 3 mm	12	N52 8x3 magnet	105
N52 magnets 5 $\times$ 2 mm	2	N52 5x2 magnet	35
NBR O-ring ~53 $\times$ 3 mm	1	o-ring 53x3	35
Silicone grease (plumbing, not auto)	1	silicone grease plumbing	70
PET film 0.3 mm A4	10 sh.	PET sheet 0.3mm A4	70
M3: 16 mm $\times$ 2, 10 mm $\times$ 3, 8 mm $\times$ 5	1 kit	M3 screw kit	70
Fishing sinkers ~40 g	1	fishing sinkers split shot	70
Aquarium silicone 100% clear, no fungicide	1–2	aquarium silicone sealant clear 100%	175
5-minute epoxy	1	epoxy glue 5 minutes	70
CA glue + thermal pad + foam + nail varnish	1	super glue CA / thermal pad servo	—

Filament for the A1

Filament	~mass	Use
PETG	~450 g	capsule, stator, cam, ring, axle, rotors, barrettes
Transparent PETG	~95 g	float dome 13e / disk 13f / plug 13g
TPU 95A	~60 g	tentacle paddles $\times$ 8 (legacy_21_flanc... file 21)

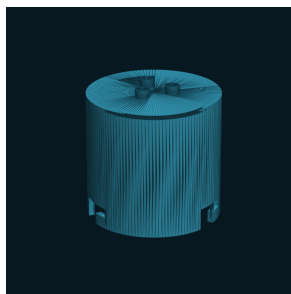
Do not buy for this first swim

Furitek Mini-Komodo (wrong drive). WS2812 LED ring. SG90 pendulum. Qi wireless TX/RX. Silicone tentacle molds 22_moule_*. Priced rows above $\approx$ 1060 THB + servo + filament + CA / pad.

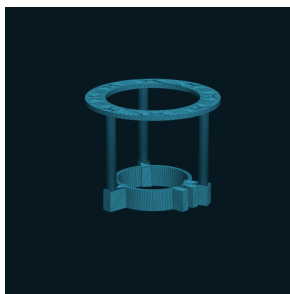
Tools. PH1 screwdriver. Cutter + metal rule. Basic soldering iron. USB-C data cable (charge-only cables will not flash). Optional: multimeter for the 5.0 V set.

Bambu Lab A1 print plan

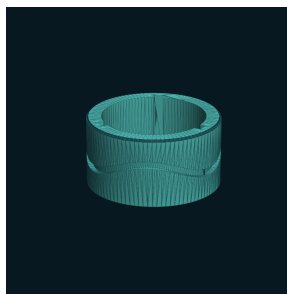
Printer: Bambu Lab A1. Bed 256 × 256 × 256 mm. Nozzle 0.4 mm. A1 is direct drive, so TPU 95A at 30 mm/s is fine. There is no kit slicer profile. Use Bambu Studio Generic PETG and Generic TPU. Do not invent PLA.



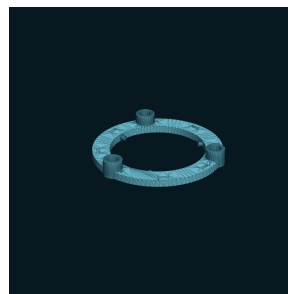
01 capsule



08 stator



06 3-lobe cam



09 sliding ring

STL previews from the kit meshes. Identify parts. These are not photos.

Magnet-test parts first — then stop

Print 01, 03, 04, 05, 07 only. Glue magnets. Test through 3 mm cardboard. Then print the rest. Do not queue the whole job overnight.

Suggested plates

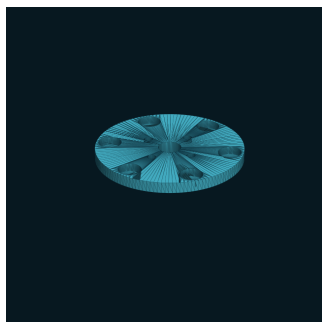
Plate	Files	Material	Notes
1 magnet test	01_capsule_upper · 03_stub_axle · 04_rotor_inner · 05_rotor_outer · 07_top_cap	PETG	Print, then STOP
2 frame	08_stator · 09_slider_ring	PETG	126 mm + 113 mm just fit
3 body + drive	02_fuselage · 06_cam_3lobe · 18_ballast_trellis · 20_tentacle_bar ×8	PETG	Fuselage is 147 mm tall
4 float dome	13_float_dome	Transparent PETG	160 mm — one part
5 float extras	13_float_disk · 13_float_plug	Transparent PETG	Disk is 145 mm
6 tentacles	legacy_21_flanc (file 21) ×8	TPU 95A	22×209 mm. 30 mm/s max

Orientation and settings (v6.2 notice)

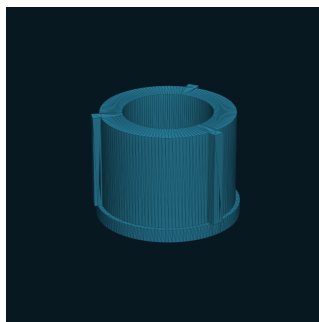
File	On the bed	Settings
01 capsule	Upside down, top face on the bed	40 % infill, 4 walls, no support
02 fuselage	Collar on the bed, tail in the air	4 walls, no support
03 axle, 07 cap	Flat	100 % infill
04 inner rotor	Magnet holes facing the bed	50 % infill
05 outer rotor, 06 cam	Upright, large opening up	3–4 walls, no support
08 stator	Upright	Tree Slim supports under the ring only
09 sliding ring	Flat, bosses up	3 walls
19 TPU paddle (file 21) ×8	Flat	TPU 95A, 30 mm/s max
20 tentacle bar ×8	Flat, teton up	PETG, one plate
13e / 13f / 13g	Skirt on bed / flat	Transparent PETG, no support
18 ballast trellis	Bucket on the bed	4 walls

Supports: none, except the stator tree-slim under the ring. That is the only support in this kit. Do not print silicone molds or old clamp/rib/paddle STLs. The paddle you want is optional/legacy_21_flanc_TPU_NOT_piece19.stl (file 21). Skip 19_tentacle_skeleton_TPU_x8.stl for this first swim — different mesh.

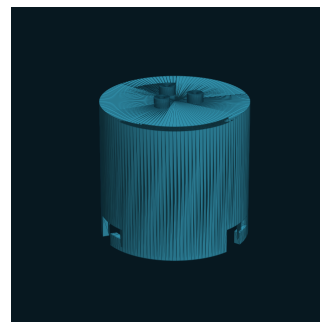
Magnet test first



04 inner rotor



05 outer rotor



01 capsule

Print plate 1. Do not queue plates 2–6 yet. Glue six 8×3 mm magnets into 04 (step 1), alternate N/S. Screw 04 onto the servo horn (step 2). Glue six 8×3 mm magnets into 05 in the opposite pattern, plus two 5×2 mm Hall pips (step 3). Varnish the wet side. Bolt axle 03 onto capsule 01 (step 4).

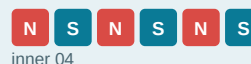
Polarity cheat

Stack 12 magnets. Mark the top of the pile. That mark is one pole.

Inner 04: mark-up, mark-down, mark-up, mark-down, mark-up, mark-down

Outer 05: invert that pattern so the two rotors ATTRACT through the wall.

Hall 5×2 pips: same way both. Polarity does not matter.



inner 04

Pass / fail

Pass: with ~3 mm of cardboard between 04 and 05, turning one rotor turns the other. They snap into alignment. They must attract, not repel. Fail: one magnet is flipped. Soak, pop, re-glue. Fix this before you spend PETG on the stator and cam.

Cut eight films

Print the gabarit PDF at 100 % — do not cut from this guide

Open docs/GABARIT_FLANCS_1-1.pdf and print it on its own at 100 % / actual size. Do not “fit to page.” The outline is 212 mm long. A scaled-down thumbnail in a build guide is waste. Do not cut films from any picture on these pages.

1. Measure the 50.0 mm control square on the gabarit paper. If it is not 50.0 mm, reprint.
 2. Tape or spray-mount the sheet on 0.3 mm PET. One A4 = two outlines = two films.
 3. Cut the solid line with a cutter and a metal rule. Keep the 12×12 mm teton at the root.
 4. Before peeling, mark dashed B1–B6 with a fine pen (placement guide along the TPU spine).
 5. Repeat until you have 8 identical films. Not 16. (The old notice title said 16; the body text and gabarit say eight.)
- Fold test: a film folded in half must spring back with no white crease. A white crease is cheap PS, not PET.

Control square (print the GABARIT at 100%)



Assembly 1–7

Have CA, silicone grease, PH1, and the varnished magnets ready. Short sentences. Check each box.

1 Inner rotor magnets · part 04

- Lay 04 flat, six round holes toward you.
- One CA drop per hole. Push each 8×3 mm magnet home so it is flush, not proud.
- Alternate N / S. A magnet glued the wrong way means the robot will not drive.

Check. A paperclip sticks to every magnet. Nothing sticks up.

2 Screw 04 onto the servo

- Clip the round servo horn onto the servo shaft.
- Set 04 on it, magnets up (away from the servo).
- Four small screws through the rotor into the horn. Spin by hand. It must run true.

Check. Hold the rotor, turn the servo body. It must not unscrew.

3 Outer rotor magnets · part 05 · this side lives in water

- Flip 05. The six magnet holes sit under the disk.
- Glue six 8×3 mm magnets, inverse of 04.
- Glue both 5×2 mm magnets in the two small edge pockets (Hall pips). Same way for both.
- Test before the CA dies: 04 and 05 face to face with 3 mm cardboard. They must attract and self-align.
- Varnish or thin-epoxy every magnet on this rotor. Nickel plating corrodes in water.

Check. Turn one rotor; the other follows through the card.

4 Axle on the capsule · 01 + 03

- Stand the capsule, top toward you. Three drilled bosses.
- Axle flange matches. Three M3 × 10. Snug, do not strip.
- Silicone grease on the whole axle stem and the flat thrust ring.

Check. The axle is vertical. It does not wiggle.

5 Stator frame · part 08

- The stator only goes on from the top. It will not pass the fuselage end.
- Slide until the split collar sits 10 mm below the capsule shoulder.
- Two M3 × 16 in the collar ears. Cinch. The frame must not rotate or slide by hand.

Check. Three round columns are vertical. Three arms join the collar to the columns (the v6 fix).

6 Servo into the can

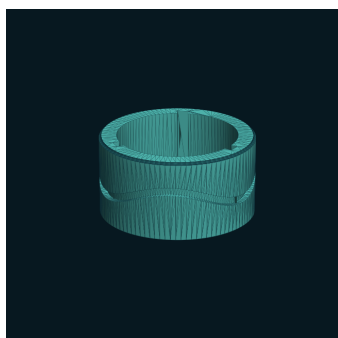
- Lower the servo + inner rotor through the bottom opening, rotor up, until ~1 mm under the thin magnet roof.
- CA the NTC onto the servo metal side. That side against the capsule wall with a thermal pad. Foam on the other side. Heat leaves into the water.
- CA the Hall sensor flat under the roof, 28 mm from centre (just under the edge of the thin zone). Three wires drop to the ESP32.
- Servo leads hang down for step 11.

Check. Spin the outer rotor on top (not yet fitted). The inner rotor follows through the wall.

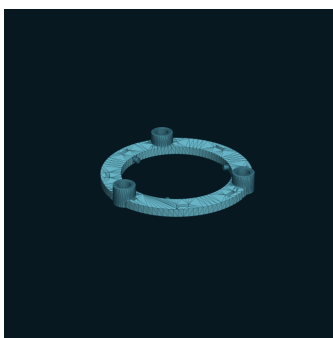
7 Outer rotor on the axle

- Grease the centre hole of 05. Slide it on, magnets down.
- It will clack onto the inner magnets. That is correct.
- Spin. It must ride the thrust ring and not scrape the capsule.

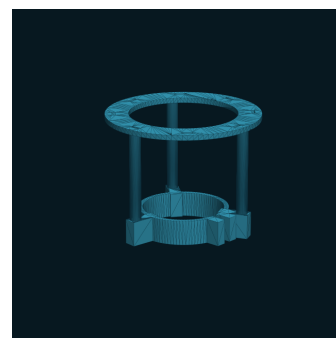
Check. One finger turns it. You feel the inner rotor follow.



06 cam — wavy groove outside



09 ring — pins ride the groove



08 stator — three columns

8 Three-lobe cam + sliding ring

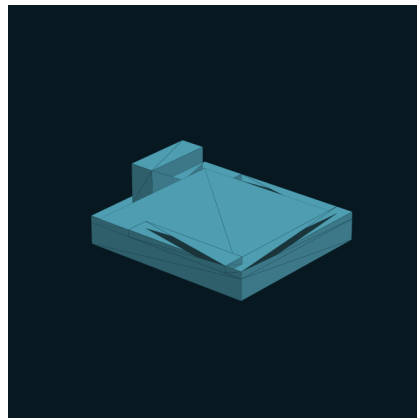
- Look at the cam: three straight inner keys, a wavy 3-lobe groove outside. Slide it onto the three tabs of the outer rotor, all the way down.
- Grease the wavy groove and the three stator columns.
- Present the sliding ring bosses up: three holes on the three columns, three pins pointing in.
- Lower it while turning the cam by hand so the pins enter the groove.
- Cap 07 on the axle. One M3 × 8 in the centre.
- Hard spot? A burr in the groove. Wrap 240-grit on a toothpick, then re-grease.

Check. Turning the cam by hand, the ring rises and falls 10 mm, stays flat, three times per turn, no force.

Tentacles · steps 9–10



TPU paddle · file 21 · print ×8



20 PETG barrette · print ×8

Hang map — do not mix these up

Barrette teton → stator (fixed ring, slots 8.6 × 3.0). TPU teton → sliding ring. Film teton → barrette cuvette.

9 Glue eight tentacles on the table

Each tentacle = 1 TPU paddle (19) + 1 PET film + 1 PETG barrette (20). This paddle has a spine and a fishtail. It has no Fin-Ray pads. Do not hunt for fourteen pads.

- Lay the TPU paddle flat, teton up.
- A thin CA line on the spine only. Not a flood.
- Press the film on in one go, centred on the spine. B1–B6 are a placement guide. Palm for 10 seconds.
- Glue the film's 12 × 12 mm teton into the barrette cuvette (a blob of CA, 10 s).
- Do all eight. No ribs to thread.

Check. Push the mid-span. It should curve toward your finger and spring back. If it is a board, too much glue hit the spine. Peel the PET with a cutter, clean, glue a thinner line.

10 Hang tentacles on both rings

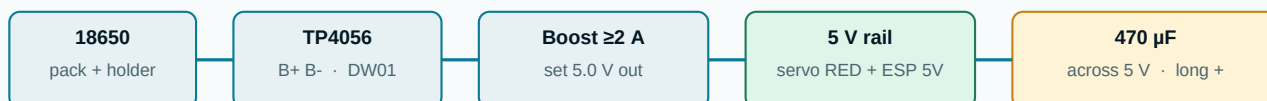
- Fixed ring (top of the stator): barrette teton from below into each of the eight slots. A drop of CA. Films now hang from the frame.
- Turn the cam until the sliding ring is LOW.
- Sliding ring: TPU teton from below into its eight slots. A drop of CA (TPU holds better with it).
- Order: eight barrettes first, ring low, then eight paddles.
- No film rubbing a column while the CA sets.
- A tentacle that beats out of time usually has its barrette and TPU swapped.

Check. Turn the cam by hand. All eight curve inward, then release, together. That is the pulse. Root play of 2 mm eats half the stroke. Push tetons fully home.

Wiring · step 11 · leak on GPIO4

GPIO4, not GPIO5

The French notice said leak on GPIO5. That pin has no ADC on the ESP32-C3. Kit firmware v6 reads leak on GPIO4. Two bare wires at the lowest point of the capsule, plus 470k from GPIO4 to 3.3 V. Analog wet threshold < 1200. Leave ENABLE_LED 0 — a WS2812 ring would steal GPIO4.



Function	Pin	Notes
Servo yellow	GPIO10	50 Hz PWM, 1500 μs stop
Reed / ILS	GPIO3	to GND. Tap run/stop. 3 s sleep
NTC 10k	GPIO0	NTC to GND, 10k to 3.3 V
Vbat divider	GPIO1	100k / 100k from pack +
Hall SS49E	GPIO2	3.3 V / GND / signal
LEAK □	GPIO4	2 bare wires + 470k to 3.3 V — NOT GPIO5

All blacks / browns together = GND. Servo current from the boost — never through the ESP32.

- Solder pack +/- to TP4056 B+ / B-. TP4056 OUT to boost IN. Set boost to 5.0 V.
- Boost 5 V out → servo red and ESP32 5V pin. All blacks together.
- 470 μF across that 5 V rail, long leg on +.
- Park the pack against the wall. CA the reed on the inner wall so you can swipe the magnet from outside.

Check after flash. Servo runs 8 s then pauses 4 s (AUTO at 42 %). Phone sees Wi-Fi EXO-MEDUSA / password medusa123. Page <http://192.168.4.1>.

Flash firmware · step 11b

Board: ESP32C3 Dev Module. Tools → USB CDC On Boot = Enabled. File: firmware/firmware_medusa.ino.

- Install Arduino IDE from arduino.cc.
- File → Preferences → Additional boards manager URLs:
https://espressif.github.io/arduino-esp32/package_esp32_index.json
- Boards Manager: esp32 by Espressif Systems (3.x is fine; the sketch also speaks 2.x LEDC).
- Tools → Board → esp32 → ESP32C3 Dev Module. USB CDC On Boot = Enabled.
- USB-C data cable. Pick the port. If none: hold BOOT while plugging in, then release.
- Open the sketch → Upload. Success looks like “Hash of data verified” / “Done uploading”. Failed to connect: BOOT trick again.

Leave flash size, frequency, partition and upload speed on Arduino defaults (typically 80 MHz, QIO, 4 MB, 921600). SuperMini is usually 4 MB. No extra library for tank swim.

Phone remote · step 11c

The jellyfish is its own Wi-Fi access point. Your phone browser is the remote. No app. There is no internet on this network. That is normal. Stay on it.

SoftAP

EXO-MEDUSA / medusa123 → <http://192.168.4.1>

iPhone (Safari)

- Settings → Wi-Fi → join EXO-MEDUSA, password medusa123.
- iOS may pop a “Log in” captive-portal page. That is the remote. Do not tap Cancel / Use without internet if you want the page to stay.
- If nothing pops: Safari, type <http://192.168.4.1> (type it, do not Google it).
- Toggle Wi-Fi if needed. Turn off VPN / iCloud Private Relay on this network.
- Optional: Share → Add to Home Screen. Still just the browser.

Android (Chrome)

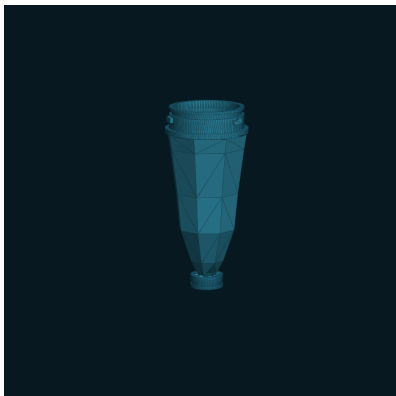
- Join EXO-MEDUSA, password medusa123.
- “Connected without internet” is correct. Stay. Disable auto-switch to mobile data if the phone hops away.
- A “Sign in to network” banner may appear — open it. Else Chrome → <http://192.168.4.1>.
- If Chrome searches: type `http://` in front of the address.

Controls

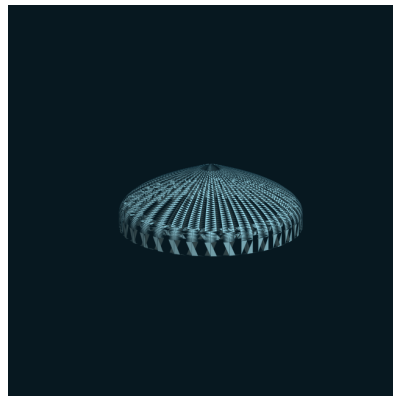
Control	What it does
SWIM	Forward 55 % (remote; auto resumes after 120 s idle)
STOP	Speed 0
AUTO	Built-in 8 s swim / 4 s glide at 42 %
REV	Reverse nudge (−35 %)
Slider	−100 % ... +100 %
RESET FAULT	Clears a fault if the leak is gone

Tiles (temp °C, battery V, RPM, mode, fault) poll /status every 1 s. BLE backup: Android Serial Bluetooth Terminal (Bluetooth LE tab) or iPhone BLE Terminal. Device EXO-MEDUSA. Text: V50, V-40, S, A, ?, R. Last order wins. The C3 is BLE only — it will never appear in the headphone Bluetooth list. If Wi-Fi + BLE together make the board flaky: `#define ENABLE_BLE 0` and re-flash.

Seal, float, tests · steps 12–14



02 fuselage · O-ring in the plug groove



13e float dome · transparent PETG

12 O-ring and fuselage

- Stretch the 53 × 3 mm NBR O-ring into the groove at the end of the fuselage plug.
- Grease it generously. Silicone grease is the seal.
- Electronics go in through the fuselage opening: battery at the bottom, boards along the wall.
- Present fuselage under the capsule: three lugs facing three vertical slots. Push home, then a sixth of a turn. Locked.
- Never force it. Resistance = a wire on the joint.

Check. It does not unwind by itself. The shoulder sits flush.

13 Float dome + ballast trellis

- The dome is the float. Its chamber must be watertight (~195 cm³).
- A continuous bead of aquarium silicone (no fungicide) on the inner shoulder. Press disk 13f on. Smooth with a finger.
- Silicone the apex plug 13g into the top hole. Wait 24 hours.
- Float test: hold the dome alone underwater 10 minutes. Zero bubbles. One bubble = redo that spot.
- Sit the dome on the stator's top ring. The lip self-centres. Three glue dots.
- Clip the ballast trellis onto the fuselage tail teton. About 40 g of fishing sinkers in the bucket. Fine-tune in step 14.

Check. The sealed dome floats hard on its own in a bucket. That keeps the jelly head-up.

14 Three tests before a tank

Test 1 — first dry pulse. Power on. Stand the robot on a glass. All eight tentacles must curl together about once a second (8 s swim / 4 s pause at 42 % in AUTO).

Test 2 — 24 h leak. A paper towel inside the capsule. Close. Weight it. Leave the capsule 24 h at the bottom of a bucket of water. The paper must come out dry. Wet paper = do not swim. Re-grease the O-ring, twist it a quarter turn in its groove, repeat.

Test 3 — tank trim. Whole robot in water. Add or remove sinkers until it sinks very slowly, upright, dome up. Swim will make up the rest.

Swipe the reed magnet to start/stop. Hold 3 seconds for deep sleep. Then film it. You earned that.

Troubleshooting

Stall / “Rotor bloqué” / FAULT: rotor blocked

The Hall no longer sees the outer rotor. Debris in the cam groove, a follower pin jammed, or magnets too far. Firmware tries three reverse kicks, then faults. Clean the groove. Turn by hand. RESET FAULT on the phone page, or a magnet tap on the reed. If it still slips: re-glue outer magnets 0.3 mm proud.

Leak fault / “FUIITE — ouvrir et secher la capsule”

GPIO4 saw a wet analog reading (threshold < 1200). Open. Dry. Find the path (O-ring, bayonet, a wire in the groove). Re-grease. 24 h paper-towel test again. RESET FAULT only after the leak is gone. Confirm the 470k pull-up is on GPIO4, not GPIO5.

Won't join Wi-Fi / no page

Stay on EXO-MEDUSA. There is no internet — phones love to hop away. Disable “auto switch to mobile data.” Type <http://192.168.4.1>, do not search. iPhone: use the captive-portal page; skip VPN / Private Relay. If the SSID never appears: ESP32 has no 5 V, empty 18650, or a charge-only USB cable during flash. TP4056 LED should show charge/idle.

Servo twitch, will not turn = missing 470 μ F

The servo brown-outs on the 5 V rail. Fit 470 μ F 10 V across servo 5 V, long leg on +. Feed that rail from the boost, never through the ESP32. A weak pack or a boost under 2 A looks the same. Set 5.0 V.

Pulses but does not go forward

Too heavy or too light — redo trim. Reverse the speed on the phone page. Check tentacles curl inward on the down-stroke.

One tentacle out of time

Film unstuck, or barrette and TPU swapped on that station.

Floats tilted or inverts

Dome chamber leaking (bucket test: zero bubbles) or not enough tail ballast. Watertight dome up + lead down = upright.

Hard spot on the sliding ring

Burr in the cam groove or on a column. 240-grit, then silicone grease. Check the three pins are intact.

Water in the capsule

O-ring dry, pinched, or a print scar in the groove. Re-grease, quarter-turn the ring, 24 h test.

No port in Arduino IDE

Hold BOOT while plugging USB, then release. Use a data USB-C cable.

Invisible on Bluetooth

Normal in the headphone list. Use a BLE terminal app. If still nothing: ENABLE_BLE 1.

“Pause thermique” / overheat FAULT

Firmware protects the servo: derate 52 °C, pause 58 °C, fault 65 °C. Check the thermal pad to the wall. Let AUTO glide. Do not swim 100 % in a hot room.

What this kit is not

No steering. No depth sensor. No LED dome (leave ENABLE_LED 0 so leak can stay on GPIO4). No Qi lid. No Festo two-plate cam. Build the second jelly after this one swims.

First swim checklist

Tick these before the jelly goes in the tank. Short list. No surprises in the water.

☐	1. Print Magnet-test plate 01/03/04/05/07 passed through 3 mm card. Outer magnets varnished.
☐	2. Print Rest of PETG, transparent dome, TPU paddles ×8 (file 21), barrettes ×8. Stator is the only part with Tree Slim supports.
☐	3. Film GABARIT_FLANCS_1-1.pdf printed at 100 %. Control square is 50.0 mm. Eight films, tetons intact.
☐	4. Tentacles Barrette teton → stator. TPU teton → sliding ring. Film teton → barrette cuvette. Cam-by-hand pulse inward.
☐	5. Wiring Leak on GPIO4 (not GPIO5). Servo red from the boost. 470 µF across 5 V, long leg on +.
☐	6. Flash firmware_medusa.ino on ESP32C3 Dev Module. USB CDC On Boot = Enabled. AUTO 8 s / 4 s.
☐	7. Phone Wi-Fi EXO-MEDUSA / medusa123 → http://192.168.4.1 . SWIM / STOP / AUTO work.
☐	8. Dry pulse On a glass, eight tentacles curl together. No stall fault. No twitch (if twitch: missing 470 µF).
☐	9. Dome Silicone, no fungicide. 24 h cure. 10 min underwater, zero bubbles.
☐	10. Capsule Paper towel inside. 24 h at the bottom of a bucket. Paper comes out dry.
☐	11. Trim Sinks very slowly, dome up, ~40 g in the tail bucket. Then swim.
☐	12. Kids / pets N52 magnets boxed. 18650 in its holder. CA capped.

EXO-MEDUSA consumer kit v6 · Bambu Lab A1 · tank-swim

STLs and firmware were not changed to make this guide.